

Qatar-4b

R-2640

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_190206 · created 2026-08-29T01:14:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458520.59754 +/- 0.00085 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.053
Transit depth (Rp/Rs)^2	3.37 +/- 1.96 [%]
Orbital Inclination (inc)	85.58 +/- 2.14
Airmass coefficient 1 (a1)	1.16 +/- 0.66
Airmass coefficient 2 (a2)	-0.05 +/- 0.19
Scatter in the residuals of the lightcurve fit is	59.38 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	6.75
Transit Duration (day)	0.0432 +/- 0.0076

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.053 vs 0.138 (deviation 0.71 σ)
Duration fit vs published	0.0432 d vs 0.0892 d (deviation 4.92 σ)
Mid-time O-C	-0.8 min
Depth SNR	2.8
Residual scatter	59.38 %

Light curve

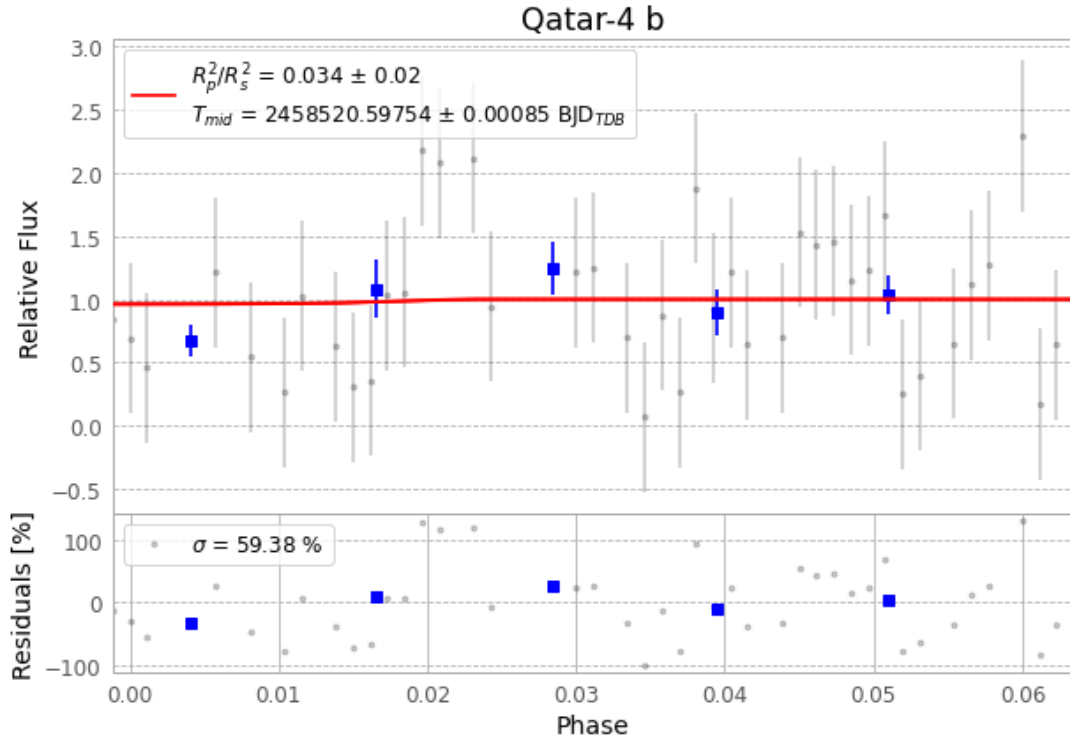


Figure 1 — FinalLightCurve_Qatar-4 b_2019-02-05.png (SHA-256 88d23d5aecccb5d3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 134], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d38ebee39bac4ee8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

58 FITS frames (38 MB), Qatar-4190206021812.FITS → Qatar-4190206050912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-190206.log (78ed8fb0a774...), FinalLightCurve_Qatar-4 b_2019-02-05.png (88d23d5aeccb...), FinalLightCurve_Qatar-4 b_2019-02-05.csv (ea7fbf43e54b...)

Compute resources

Wall time 158.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 01:14Z.